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Introduction to Computing

Title : **Internet of Things (IoT) and Smart Cities**

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Abstract

The burgeoning proliferation of the Internet of Things (IoT) is a key catalyst in the advancement of Smart City initiatives globally. Everyday objects are increasingly being integrated with electronic components and network protocols, facilitating their interconnectivity and access to the Internet. A forecast by Gartner indicates that by 2020, smart cities will incorporate approximately 50 billion connected devices. These devices are instrumental in transforming urban environments into 'smart' cities, yet they also bring forth potential security vulnerabilities and privacy concerns. Recent developments in smart city projects have highlighted both the anticipated benefits and the inherent risks. This paper explores the current and prospective landscape of smart cities and IoT. It delves into the synergy between smart cities and IoT, shedding light on the factors driving the growth and evolution of these technologies. Additionally, the paper addresses the vulnerabilities present in IoT frameworks, particularly in the context of smart cities, and proposes strategies to mitigate these issues.

Key Words: Internet of Things (IoT); Smart Cities, IoT Technology, IoT Challenges

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1 Introduction

In the face of urban expansion, the adoption of smart and innovative solutions becomes essential to enhance productivity, improve operational efficiency, and minimize management expenses [1]. The trend of integrating homes with IoT devices like TVs and Internet boxes is steadily rising. In the realm of real estate, the use of connected devices encompasses thermostats, smart alarms, intelligent door locks, and various other systems and appliances. The 2016 United Nations Cop21 climate change conference in Paris notably discussed the role of connected devices, offering numerous communities the chance to revamp their environmental goals by leveraging IoT technologies to curtail CO2 emissions*†. IoT's significance in smart city frameworks is increasingly evident. For instance, smart waste containers can inform when they are nearing capacity, prompting timely emptying. Residents can ascertain the status of street waste containers through smartphone applications. Additionally, this data enables companies to optimize garbage collection routes. Sensor-equipped locales can monitor environmental conditions, aiding cyclists or athletes in choosing the healthiest routes, while the city can make traffic adjustments or increase greenery in certain areas. This openly accessible data fosters the development of applications using real-time information for residents, making cities centers of knowledge sharing.

As we delve into the age of smart cities, the necessary technologies and solutions are just beginning to surface. Figure 1 illustrates a smart city example. According to Gartner [2], IoT investment is critical for the development of smart cities, with data utilization services anticipated to drive most revenue. Home safety and security are projected to be the second-largest market in service revenues. The health and well-being sector is expected to reach a market value of $38 billion by 2020 [2]. Balancing effectiveness and privacy concerns is crucial; a sophisticated attacker could potentially hijack various smart devices such as lights, cameras, traffic signals, and connected vehicles. With an estimated 50 billion devices connected by 2020, municipal safety in intelligent cities becomes a paramount concern [2][3]. Addressing the safety, security, and privacy challenges in smart cities requires a multifaceted approach that spans technology, sociology, legal frameworks, and policy management.

2 Internet of Things and Smart City

Recent studies have explored various interpretations of the Internet of Things (IoT), with several authors contributing definitions to the field [4][5][6]. One conceptualization of IoT describes it as a network of objects possessing unique identities and virtual personas, operating within smart environments and employing intelligent interfaces for interaction across diverse contexts such as social, medical, environmental, and user-related domains [7]. The IoT sector is witnessing substantial financial investments to facilitate a broad spectrum of services. This financial commitment is evident in the trust shown by investors, highlighted by significant investments from major corporations and industry groups. Notable examples include Telefónica, SK Telecom, NTT Docomo Ventures, Elliott Management Corporation, GDF SUEZ, and Air Liquide, all of which have allocated substantial funds towards IoT research and development.

The effective deployment of IoT relies on the establishment of communication standards that ensure seamless interoperability among diverse objects. Several global organizations are at the forefront of standardizing these communications, including the International Telecommunication Union (ITU), the Institute of Electrical and Electronics Engineers (IEEE), the Internet Engineering Task Force (IETF), Global Standard1 (GS1), the Organization for the Advancement of Structured Information Standards (OASIS), and the Industrial Internet Consortium (IIC), among others. A summary of some key IoT standards and initiatives is presented in Table 1. For instance, the ITU's Internet of Things Standard Global Initiative (IoT-GSI) has introduced important recommendations like the ITU-T Y.2060 [8], which outlines the IoT concept, and the ITU-T Y.2061 [8], detailing requirements for machine-oriented interfaces.

Various standards have been proposed by organizations such as the IEEE and IETF, addressing different layers of sensor networks based on the Internet Protocol (IP). For example, the IEEE 802.15.4 standard, preferred over Ethernet in industrial settings, operates at the link layer. Meanwhile, at the network layer, the 6LoWPAN (IPv6 over Low power Wireless Personal Area Networks) standard adapts IPv6 protocol for wireless communication [9]. Furthermore, the IETF's publication of the IPv6 Routing Protocol (RPL) standard in 2011 marked a significant advancement for Low-power Networks.

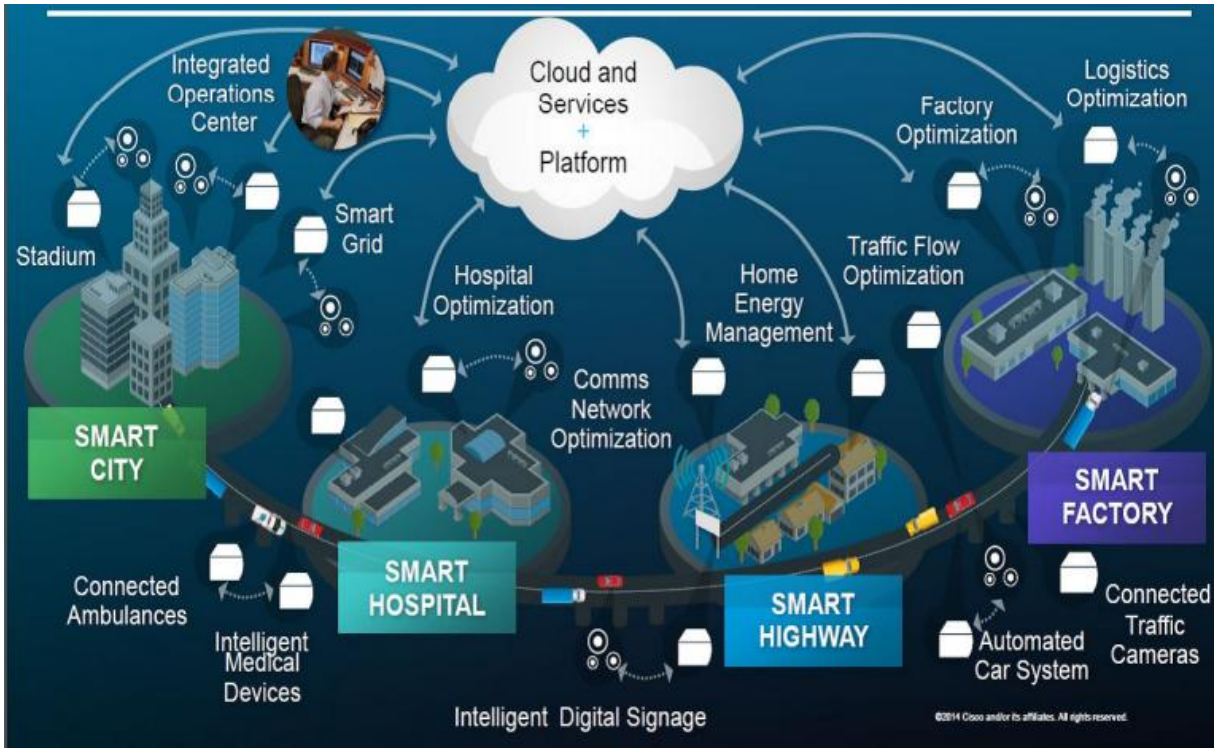


Figure 1 Development of smart cities

The Internet Engineering Task Force (IETF) has taken significant strides in standardizing protocols for connected objects, particularly focusing on an application layer-oriented protocol. This led to the development of the Constrained Application Protocol (CoAP), detailed in RFC 7252 as of June 2014. CoAP, akin to HTTP, provides various methods and commands, such as HTTP Get, to interact with an object and modify its state. It primarily utilizes UDP and can be optionally secured with Datagram Transport Layer Security (DTLS) for enhanced communication security.

The EPC Global initiative, overseen by GS1, established a unique identifier system and network architecture for electronic products [10] [11]. The Object Naming Service (ONS) in this architecture is based on the principles of the Domain Name System (DNS). The EAN standard, introduced in 1970, was initially used for product identification but only identified product categories, not individual items. To address IoT needs, the EPC standard was developed by GS1, requiring distinct IP addresses for each connection. OASIS has recommended IoT network and messaging technologies like MQTT, AMQP, and DDS. The Industrial Internet Consortium (IIC) guides IoT technologies for the Industrial Internet, with notable members including Bosch, Intel, IBM, Schneider, Huawei, and Cisco. Smart cities integrate physical, ICT, social, and business infrastructures, utilizing IoT and Wireless Sensor Networks (WSNs) for machine-to-machine and human-to-machine communications, enabling information sharing across platforms and applications [13] [14].

3 IoT Architecture

The integration of IoT technologies into expansive networks is anticipated, involving thousands of devices and extensive areas. This chapter primarily examines the LoRa Ultra-Narrow Band (UNB) technology, developed collaboratively by Semtech and SigFox.

3.1 LoRa

LoRa is a low-power, wide-area wireless communication technology for IoT applications. It offers long-range capabilities, low power consumption, and secure data transmission [15]. LoRa enables easy device interoperability without complex setup processes, finding applications in various domains like home energy management, security systems, healthcare monitoring, and transportation [15]. It operates on regional or global scales, utilizing frequencies in the 433, 868, or 915 MHz ISM bands, with modulation techniques like GFSK or its unique LoRa modulation in Europe [16]. LoRa networks have a hierarchical star-based topology, with devices functioning as servers, endpoints, or gateways [17]. It supports data rates ranging from 0.3 Kbps to 50 Kbps in Europe, with payload capacities from 2 to 255 bytes [17]. LoRa's minimal infrastructure and operational costs make it popular among wireless network operators, and it can be scaled up by adding more gateways [17]. Major telecommunications providers are adopting LoRa for nationwide networks, attracted by features like bi-directional communication, asset tracking, security, and localization capabilities [18].

3.2 SigFox

SigFox has developed a unique ultra-narrowband communication system specifically for IoT applications, capable of supporting long-range connections, such as those exceeding 20 km between a client device and a base station. Operating within the license-exempt 868 MHz band, SigFox's technology is engineered for highly efficient, low throughput communications, significantly reducing the need for antennas and base stations. In comparison to certain cellular networks, SigFox's approach requires approximately 1,000 times fewer antennas and base stations for equivalent coverage*. This system also provides a service management interface, facilitating control over key communication aspects like battery life, temperature settings, signal quality, data volume, among others. SigFox's network has successfully connected a multitude of devices across various major cities globally. Presently, it is operational in 14 countries, covering an area surpassing 1.2 million km² and serving around 223 million individuals.

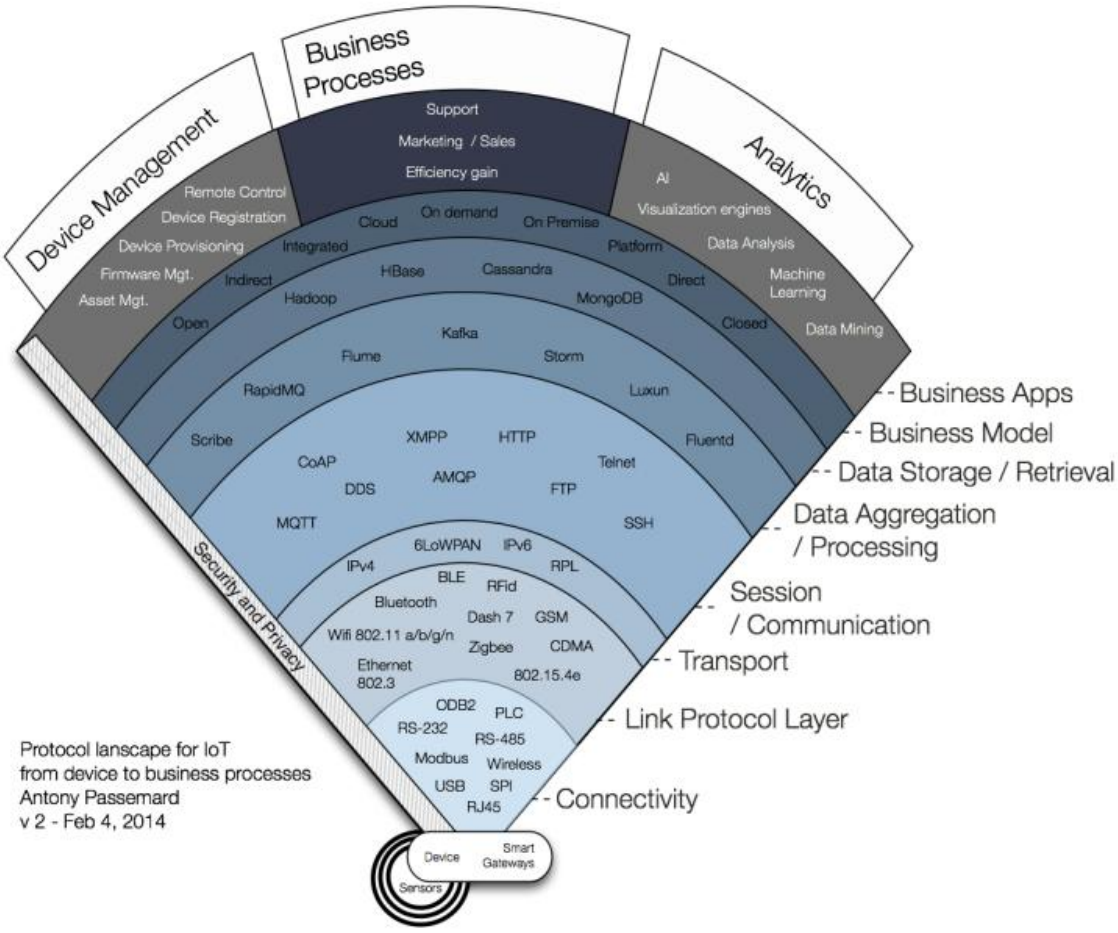


Figure 2 Internet of Things protocol stack [12]

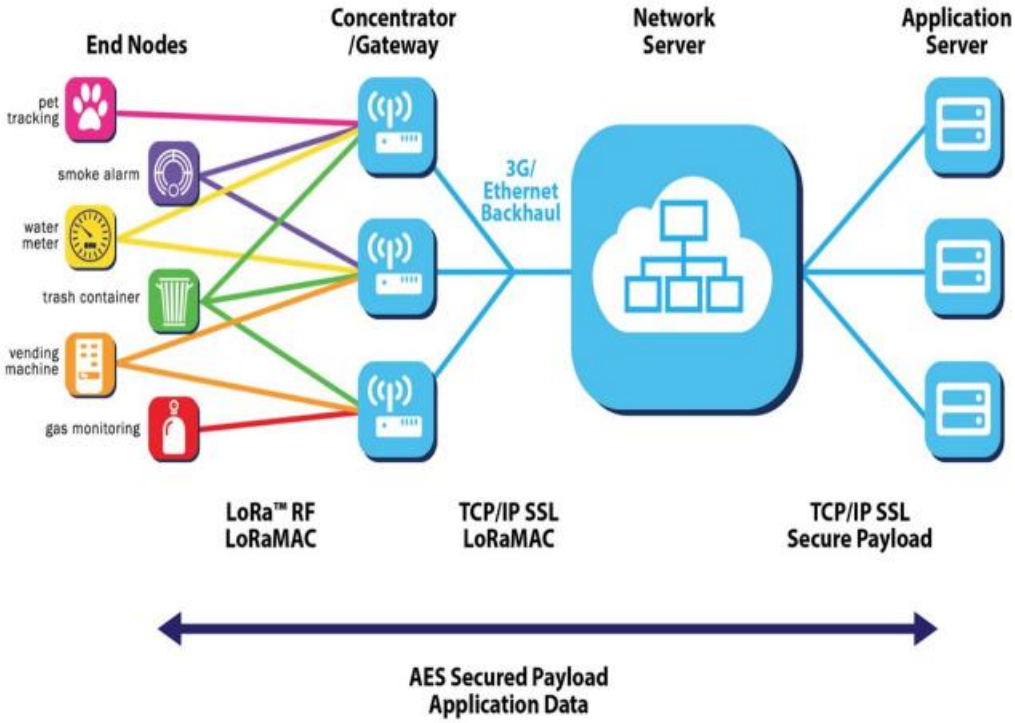


Figure 3 LoRa architecture [18]

IoT, in its broad scope, can be categorized into three primary layers: the perception layer, the network layer, and the application layer. The perception layer is responsible for capturing, collecting, distinguishing, and identifying information from objects in the physical world [5]. This layer comprises elements such as RFID tags, cameras, GPS devices, sensors, and laser scanners. The network layer primarily handles the forwarding of packets through a reliable communication medium. Finally, the application layer is involved in processing the data, integrating information from various sources, and presenting it to end-users. For a comprehensive understanding, Table 3 presents a comparison of different low-power WAN technologies applied in various IoT scenarios.

4 IoT as an Enabling Technology for Smart Cities

IoT plays a pivotal role in the evolution of Smart Cities worldwide, offering a variety of ubiquitous services. This technology opens up new possibilities, such as the ability to remotely monitor and control devices, as well as analyze and respond to data derived from diverse real-time traffic sources. Consequently, IoT innovations are transforming urban landscapes by enhancing infrastructure, fostering more efficient and cost-effective municipal services, optimizing transportation to alleviate traffic congestion, and bolstering public safety. To unlock the full potential of IoT in urban environments, those designing and implementing smart city solutions recognize the necessity of not just offering isolated smart features, but providing comprehensive, scalable, and secure IoT systems. These systems should encompass efficient IoT solutions, seamlessly integrating various aspects of urban life.

4.1 IoT for Smart Cities: Key Requirements and Practical Examples

For smart city solutions to be effective, they must integrate IoT platforms that can handle the demands of modern IoT, managing a vast network of connected devices, systems, and individuals. An ideal IoT platform should:

- Minimize costs and risks associated with developing and evolving IoT services.
- Enable connection of diverse systems within a city.
- Reduce the time needed to implement and deploy IoT services integral to smart city projects.
- Provide secure, scalable service access, opening new possibilities for urban development.
- Generate value (e.g., enhanced services) from the interconnection of smart devices and data.

Today, IoT devices with a variety of functions (like temperature, light, humidity, and pressure sensors) are increasingly common, shifting from reactive to predictive capabilities. Various sectors, including healthcare, manufacturing, and transportation, are actively deploying connected objects. IDC [19] forecasts that China's IoT market will hit $361 billion by 2020, growing by 13.3% over five years. Major Chinese companies such as Alibaba, Baidu, Huawei, Lenovo, and Xiaomi are heavily investing in IoT. The "Chinese Silicon Valley," Zhongguancun in Beijing, is becoming a hub for IoT entrepreneurs globally.

In the United States, the White House's 2015 Smart City Initiative focuses on fostering technological collaboration among cities, federal entities, universities, and private sectors.

Kansas City, in partnership with Sprint and Cisco, is developing North America's largest smart city project to enhance municipal services. This $15 million project includes a comprehensive sensor network and Wi-Fi to gather and share various city data with citizens.

In Japan, Fujisawa, a city being developed by Panasonic, is expected to house 3,000 people by 2018. Similarly, the construction of Songdo in South Korea by Gale, an American real estate firm, aims to attract around 300,000 workers and 65,000 residents by 2020. The UAE is investing $18 billion in Masdar, a city powered entirely by renewable energy, focusing on sustainable practices like wastewater reuse for irrigation.

Sweden's Malmo Green Digital City project aims for carbon neutrality by 2020 and complete reliance on renewable energy by 2030. This initiative plans to accommodate 10,000 residents, with an additional 20,000 people working or studying there. Innovative features in these cities include sensor-based street lighting in Fujisawa and rainwater recycling in Songdo.

In France, the Ministry of Transport launched the scoop@F project in 2016, focusing on smart vehicle infrastructure. This project, with a budget of 20 million euros, co-funded by the state, communities, industry, and the EU, involves testing 3,000 intelligent vehicles across six locations, including Ile-de-France and the Bordeaux ring road. These vehicles, connected through Wi-Fi, 4G, or 5G, will share real-time traffic information, improving road safety and driving conditions [20] [21].

To summarize, smart cities are interconnected urban environments that leverage telecommunications and information technologies to enhance the quality of life for their inhabitants. The realization of a smart city hinges on two fundamental goals:

- Developing sophisticated urban infrastructure capable of gathering and analyzing data through cutting-edge technologies such as smart grids, smart meters, intelligent buildings, connected devices, and big data. This infrastructure aims to proactively identify and address potential issues.

- Enabling residents to engage with their surroundings using intelligent applications. This interaction is designed to reduce CO2 emissions, thereby lessening environmental pollution. The resultant cleaner environment contributes to improved public health and enhances overall living conditions, including safer, more efficient, and cost-effective transportation.

4.2 IoT Applications in Smart Cities

Exploring how the IoT paradigm can transform urban environments is a topic of considerable interest. Numerous national governments are actively exploring and planning to integrate Information Communication Technology (ICT) solutions for public service

management to realize the Smart City concept [22].

4.2.1 Health of Buildings

Preserving the integrity of historical buildings within a city requires: (1) constant monitoring of their condition, and (2) identification of areas most affected by external factors [23]. Cities house a variety of structures, differing in size and age, with many being historical (like buildings, dams, or bridges [24]). To monitor building conditions, passive Wireless Sensor Networks (WSNs) can be integrated into structures, periodically transmitting signals to report on the building's health [14].

4.2.2 Smart Health

Wireless Body Area Networks (WBANs), based on low-cost WSN technology, are transforming patient monitoring in hospitals and residential settings [25]. These miniature sensors, either implanted or worn on the body, measure various physiological parameters and communicate with medical devices using WPAN technologies (ZigBee, 6LowPAN, CoAP, etc.). The data collected is then analyzed remotely (refer to Figure 5). Sensor design must prioritize wearability, ensuring they are lightweight, small, and non-intrusive. Moreover, these sensors must be energy-efficient to operate on small batteries [26].

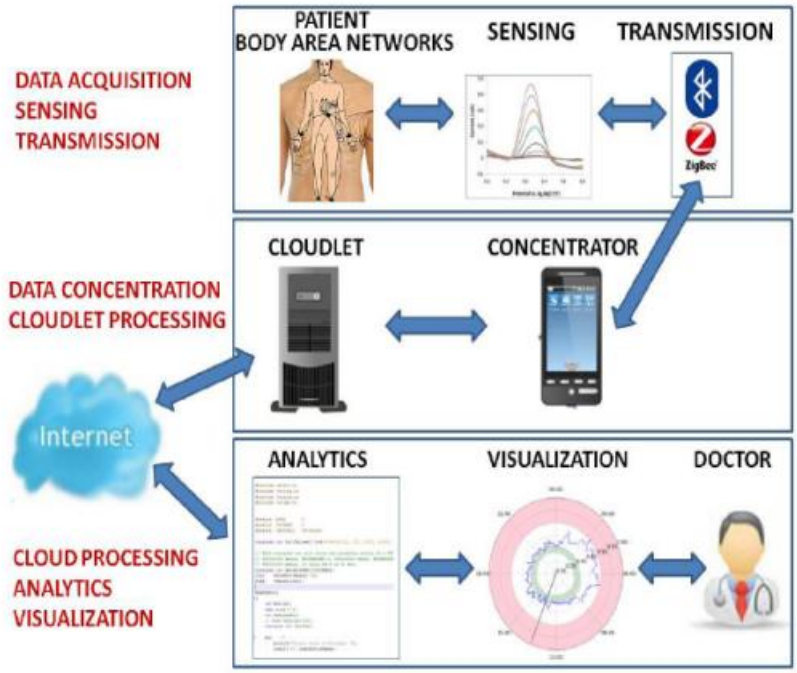


Figure 4 Components of a remote patient monitoring system that is based on an IoT-Cloud architecture [26]

4.3 IoT Challenges for Achieving Smart Cities

The multifaceted nature of IoT, involving numerous technologies, presents two primary categories of challenges: (1) issues stemming from the underlying technologies of IoT, and (2) new challenges arising from IoT implementations. Figure 8 displays the critical IoT characteristics for achieving smart cities and the hurdles to be surmounted. Central issues include scalability, networking and transport, heterogeneity, as well as privacy and authentication. This section delves into these challenges, with Table 5 encapsulating common IoT issues as per the three-layered architecture proposed by ITU-T [27].

4.3.1 Networking and Transport Issues

In IoT, a vast array of objects needs to be connected and accessible, with content retrievable by authorized users irrespective of their location. This requires efficient addressing policies. While IPv4 is currently prevalent, its limited address space is rapidly depleting, necessitating alternatives like IPv6. Efforts to integrate IPv6 with IoT, such as 6LowPAN for WSNs, are ongoing [28]. Challenges include integrating RFID tags, which use unique identifiers rather than MAC addresses as per EPC global standards [29], into IPv6 networks. Studies on merging RFID identifiers with IPv6 addresses are in progress [30][31][32], but this field lacks maturity and standardized methodologies.

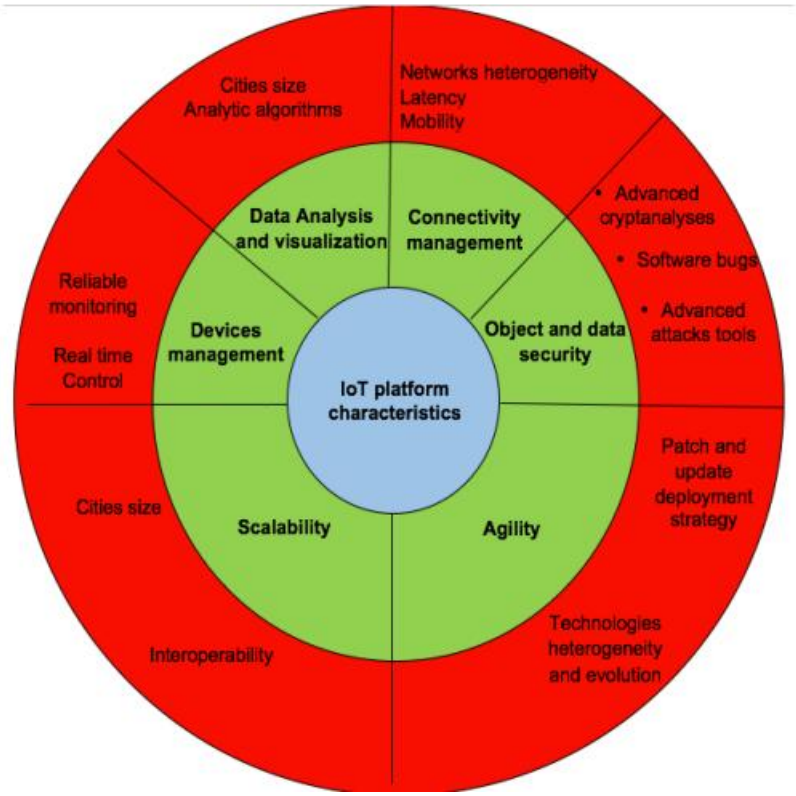


Figure 5 : IoT characteristics to achieve smart city and their limitations

Table 1 Overview on main IoT issues

	Issues	Confidentiality	Integrity	Authentication	Authorization	Unlinkability	Anonymity	Traceability	Non-Repudiation
Application Layer	Applications' issues	x	x	x	x	x	x	x	x
	Storage issues	x	x	x	x		x		
	Key management issues	x	x	x			x	x	x
	Trust management issues		x	x	x				x
	Middleware issues	x	x	x	x	x	x	x	x
	Data integrity issues		x	x					
	Data confidentiality issues	x	x						x
	Data authentication issues		x	x					x
Transportation Layer	WIFI issues	x	x	x	x	x	x	x	x
	ZigBee issues	x	x	x	x	x	x	x	x
	3G issues	x	x	x	x	x	x	x	x
	GPRS issues	x	x	x	x	x	x	x	x
	Network access issues		x	x	x	x	x	x	
	Routing protocols issues		x	x			x		
	Encoding issues		x						
	Heterogeneity issues		x	x			x	x	x
Perception Layer	RFID issues	x	x	x	x	x	x	x	x
	WSN issues	x	x	x	x	x	x	x	x
	RSN issues	x	x	x	x	x	x	x	x
	GPS issues	x	x	x	x	x	x	x	x
	Platform issues	x	x	x	x	x	x	x	x

DNS resolves IP addresses in traditional networks, but IoT demands a unique Object Name Service (ONS) [33][34][5], especially for RFID tags. ONS should link tag identifiers or IP addresses to corresponding URLs or object descriptions, a design still under research [5].

Transport layer objectives include ensuring end-to-end reliability and managing congestion. Traditional networks use TCP, but it's unsuitable for IoT due to several reasons [5][35]:

1. Connection Setup: TCP's three-way handshake is resource-intensive for IoT's typically small data exchanges.

2. Congestion Control: TCP's end-to-end congestion control doesn't align well with IoT's

predominantly wireless communications and small data transfers [36].

3. Data Buffering: TCP requires buffer storage at both source and destination, which may be too demanding for IoT devices.

Thus, TCP isn't efficient for IoT's end-to-end transmission control, calling for new transport layer protocol solutions [9]. Secure and effective routing protocols are crucial, given the risk of attacks on this layer impacting network operations. Traditional solutions for ad hoc and sensor networks don't fully meet IoT's needs. For instance, IoT's diverse objects and scenarios make it more susceptible to DoS attacks, with potentially severe impacts. Implementing Intrusion Detection Systems (IDSs) in IoT infrastructure is complex due to the unique characteristics and capabilities of IoT objects.

Traffic characterization in IoT, with its highly heterogeneous objects, hasn't been extensively studied [5]. Understanding traffic patterns is vital for network providers to plan infrastructure expansions and develop appropriate Quality of Service (QoS) solutions.

4.3.2 Security Issues

Securing the Internet of Things (IoT) is vital for businesses and government bodies, as highlighted by the U.S. Federal Trade Commission (FTC) [37]. The widespread implementation of IoT poses security and privacy risks that must be effectively addressed. In critical IoT applications, the use of inaccurate or tampered data can have severe consequences. While traditional security measures like authentication and data integrity are essential, applying them to IoT devices can be challenging due to limited resources. These security concerns hinder the global adoption of IoT, as users are hesitant without assurances of privacy and data protection. IoT devices are particularly vulnerable to attacks due to factors such as physical accessibility, reliance on vulnerable wireless communications, and limited capacity for robust security measures. Unique vulnerabilities include diverse operating systems, lack of standard security protocols, and compromised physical security. Cryptographic methods are considered the best solution for ensuring data security in IoT [38].

1) Data Confidentiality, Integrity, and Authentication:

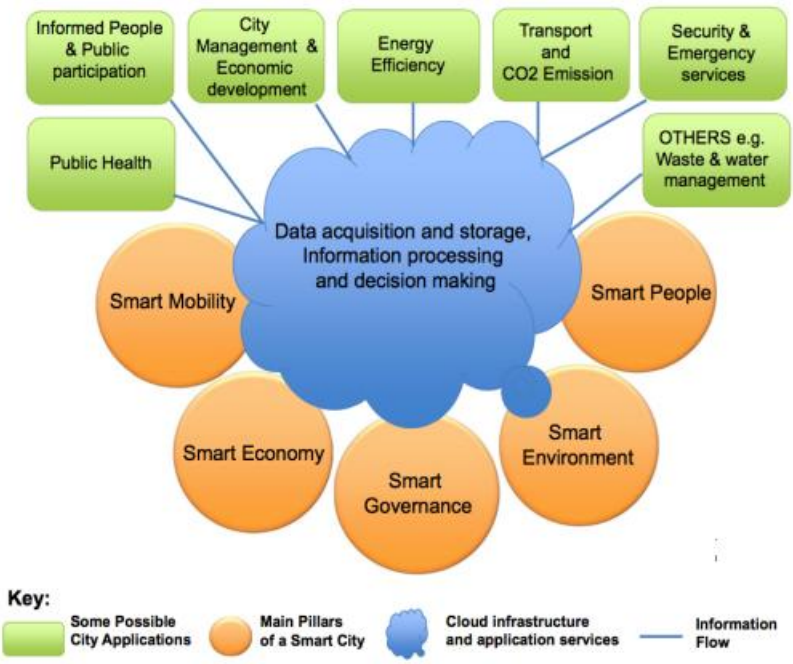
High data security, including confidentiality and integrity, is necessary in many IoT scenarios. Data encryption, categorized into symmetric and public-key algorithms, can address this. However, public-key algorithms are resource-intensive, making them less suitable for low-power devices. Symmetric algorithms, though more efficient for such devices, have limitations like complex key exchange protocols that hinder scalability [7], and risks associated with shared key confidentiality. Solutions like using different symmetric keys for different groups have been proposed, but risks remain. Public-key algorithms, including Rabin's Scheme [39], NtruEncrypt[40], and Elliptic Curve Cryptography (ECC) [41], offer

scalability without complex key management, but their application in IoT, especially for devices like RFID tags, is still being explored [38]. Trust issues also arise with public key encryption, as the base station holding public keys cannot verify the authenticity of devices.

2)Denial of service

The proliferation of internet-connected devices in urban areas presents a significant risk for cyberattacks [42] [43]. In large cities, where tens of thousands of devices are simultaneously interacting with users and each other, the security stakes are particularly high. Smart cities, with their extensive IoT networks, are prime targets for hackers looking to assemble botnets. An IoT botnet is a collection of compromised devices, often used without the knowledge of their legitimate owners, to carry out various malicious activities.

A notable example of such a threat occurred in 2016 when Dyn, a company providing DNS services, was overwhelmed by a denial of service attack. This attack was orchestrated using a vast number of internet-connected devices, primarily cameras produced by the Chinese manufacturer XiongMai. The attack's success in disrupting Dyn's services was largely due to the exploitation of Mirai malware [44]. Mirai targets vulnerabilities in connected devices, such as unchanged default passwords, turning them into tools for cyberattacks. This incident underscores the growing trend of IoT networks being exploited as platforms for malicious activities.



5 Conclusion

With the expansion and the growth of cities, making them smart becomes vital. Indeed, numerous governments such as US, Chinese or UAE launched smart cities' projects, e.g. Malmo, Fujisawa, Songdo and Masdar.

IoT represents the best way to make a city smart. Indeed, IoT can applied in multiple scenarios such as monitoring of building's status with passive WSNs, environmental monitoring e.g. Gas concentration, Water level for lakes or soil humidity, waste management, smart parking, reducing CO2 footprint, or autonomous driving. Achieving such goals needs a tremendous number of connected objects. Indeed, the number of connected objects is growing exponentially and it is estimated that 50 billion connected objects will be deployed in smart cities by 2020. However, this high number will open up numerous risks and privacy issues.

In this work, we presented an overview of IoT in the context of smart cities, and discussed how it can enhance a city's smartness. We also identified the weaknesses and risks associated to IoT deployment and adoption in the smart city environment. As part of our future work, we plan to survey the different solutions and recommendations to address several of the challenges of IoT and smart cities we have discussed in this paper and in particular the security challenges and issues.

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